

INTERESTS

I am a **Speech** and **Multimodal AI** Researcher. I build intelligent models for **Listening** (speech enhancement 🗣️), **Thinking** (audio understanding 🧠), and **Speaking** (speech synthesis, speech-to-speech 🗣️), with large language models (LLMs) 🧠, videos or images 📺, and human brains 🧠. Please check 🗣️🧠🗣️🧠🗣️🧠 symbols in publications.

SCHOOLS

Columbia University New York, NY, US
Ph.D. Candidate after joint M.S. in Electrical Engineering 2022.09 - 2027 (expected)

- Advisor: Prof. Nima Mesgarani
- Research: Speech AI, Auditory Perception, Multimodal LLMs
- GPA: 4.115/4.00

University of Illinois Urbana-Champaign Urbana, IL, US
B.S. in Computer Engineering with the Bronze Tablet 2018.08 - 2021.12

- Advisor: Prof. Paris Smaragdis
- Research: Sound Separation
- Minor in mathematics
- GPA: 4.00/4.00

Selected Courses: *Speech & Audio Processing & Recognition, Machine Learning Frontier, Machine Learning for Signal Processing, Natural Language Processing, Computer Vision, Deep Learning, Algorithms & Models of Computation, Data Structures, Database Systems*

INTERNS

Meta Superintelligence Labs (MSL) | Menlo Park, CA 2026.05 - 2026.09

Research Scientist Intern, Voice Modeling

Mentors: *Osama Hanna, Shun Zhang*

Project: *Conversational Speech Aesthetic Model with Reinforcement Learning from Human Listeners* (paper soon)

Microsoft Research (MSR) | Redmond, WA 2025.05 - 2025.08

Research Intern, Audio and Acoustics

Mentors: *Sebastian Braun, Hannes Gamper*

Project: *Sci-Phi: A Large Language Model Spatial Audio Descriptor* [pdf]

Amazon Search | Palo Alto, CA (remote) 2022.05 - 2022.07

SDE Intern, Search Science and AI & 2021.05 - 2021.08

Search data quality metrics and deep ranking model distributed training.

National Center for Supercomputing Applications (NCSA) | Urbana, IL 2020.06 - 2020.07

Student Research Intern, GPU Computing for Bionanotechnology Simulation

GPU/CUDA implementation of fast cluster-based biological molecular simulation.

SCHOOL SERVICES

Teaching Assistant | New York, NY 2025.01 - 2025.05

Speech&Audio Proc&Rec (ELENE6820)

Designed a programming assignment on spoken dialog system (cascaded and speech LM).

Teaching Assistant | New York, NY 2023.01 - 2023.05

Speech&Audio Proc&Rec (ELENE6820)

Designed a programming assignment on speech recognition and self-supervised learning.

Lab Assistant | Urbana, IL 2019.09 - 2019.12

Introduction to electronics (ECE110), Honors Section

Tutored students on programming, embedded devices, and machine learning.

Jiang, X. denotes equal contributions; [†] denotes mentored students.

Conversational Voice Aesthetic Model with Reinforcement Learning from Human Listeners
Jiang, X., Zhang, S., Jayashankar, T., Li, Y. A., and Hanna, O. (2026). (*in review*),  

[C19] **Cocktail-Talker: Multi-Speaker Dialog Modeling in Noisy Social Environments with Turn Action GRPO**

Jiang, X., Shimizu, R., Dindar, S. S., Wu, J., Xu, Z., and Mesgarani, N. (2026). *2026 IEEE Spoken Language Technology Workshop (SLT)*, [\[pdf\]](#) [\[code\]](#)   

[J5] **Video Generation Models: A Survey of Post-Training and Alignment**

Li, C., Gu, X., Kulkarni, Y., Im, E. W., Honarmand, M., Wang, Z., Song, J., Du, F., Jiang, X., Zheng, K., Li, T., Tao, F., and Fazli, P. (2026). *Transactions on Machine Learning Research (TMLR)*, [\[pdf\]](#) [\[code\]](#)   

Sympatheia: Emotionally Adaptive Voice Assistant with Continuous Affect Conditioning

Dindar, S. S., Shimizu, R., Jiang, X., and Mesgarani, N. (2026). (*in review*), [\[pdf\]](#)  

[C18] **AVMeme Exam: A Multimodal Multilingual Multicultural Benchmark for LLMs' Contextual and Cultural Knowledge and Thinking**

Jiang, X., Wang, Q., Wu, J., He, X., Xu, Z., Ma, Y., Piao, M., Yang, K., Zheng, X., Shimizu, R., Chen, Y., Firoozi, A., Mischler, G., Dindar, S. S., Antonello, R., He, L., Hsieh, T.-A., Fan, X., Wu, Y., Ma, Y., Amballa, C., Chen, W., Hai, J., Li, R., Choudhari, V., Han, C., Li, Y. A., Flinker, A., Elhilali, M., Benetos, E., Hasegawa-Johnson, M. A., Roy Choudhury, R., and Mesgarani, N. (2026). *2026 Conference on Language Modeling (COLM)*, [\[pdf\]](#) [\[data\]](#)   

[C17] **MeanFlow-TSE: One-Step Generative Target Speaker Extraction with Mean Flow**

Shimizu, R., Jiang, X., and Mesgarani, N. (2025). *2026 INTERSPEECH*, [\[pdf\]](#) [\[code\]](#) 

Speaker Identity is Robustly Encoded in Spatial Patterns of Intracranial EEG for Attention Decoding

Dindar, S. S., Jiang, X., Choudhari, V., Bickel, S., Mehta, A., Schevon, C., McKhann, G. M., Friedman, D., Flinker, A., and Mesgarani, N. (2025). [\[pdf\]](#)  

[J4] **Sci-Phi: A Large Language Model Spatial Audio Descriptor**

Jiang, X., Gamper, H., and Braun, S. (2025). *IEEE Open Journal of Signal Processing (OJSP) ICASSP track*, [\[pdf\]](#)  

[C16] **SightSound-R1: Cross-Modal Reasoning Distillation from Vision to Audio Language Models**

Wang, Q.[†], Jiang, X., He, L., Wu, J., and Mesgarani, N. (2025). *2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, [\[pdf\]](#)   

[C15] **Bridging Ears and Eyes: Analyzing Audio and Visual Large Language Models to Humans in Visible Sound Recognition and Reducing Their Sensory Gap via Cross-Modal Distillation**

Jiang, X., Wu, J., Choudhari, V., and Mesgarani, N. (2025). *2025 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, [\[pdf\]](#)   
BEST PAPER AWARD!

[C14] **DMOSpeech 2: Reinforcement Learning for Duration Prediction in Metric-Optimized Speech Synthesis**

Li, Y. A., Jiang, X., Tao, F., Niu, C., Xu, K., Song, J., and Mesgarani, N. (2025). *2026 The Association for the Advancement of Artificial Intelligence (AAAI)*, [\[pdf\]](#) [\[code\]](#) 

[C13] **Layer-wise Minimal Pair Probing Reveals Contextual Grammatical-Conceptual Hierarchy in Speech Representations**

He, L., Wang, Q., Jiang, X., and Mesgarani, N. (2025). *2025 Empirical Methods in Natural Language Processing (EMNLP)*, [\[pdf\]](#)  

- [C12] **AAD-LLM: Neural Attention-Driven Auditory Scene Understanding**
Jiang, X., Dindar, S. S., Choudhari, V., Bickel, S., Mehta, A., McKhann, G. M., Flinker, A., Friedman, D., and Mesgarani, N. (2025). *2025 Annual Meeting of the Association for Computational Linguistics (ACL), main conference*, [pdf] [code] 🎧 🔧 🧠 🧠
- [C11] **ArrayDPS: Unsupervised Blind Speech Separation with a Diffusion Prior**
 Xu, Z., Fan, X., Wang, Z., Jiang, X., and Choudhury, R. R. (2025). *2025 International Conference on Machine Learning (ICML)*, [pdf] 🔧
- [C10] **Exploring Finetuned Audio-LLM on Heart Murmur Features**
 Florea, A., Jiang, X., Mesgarani, N., and Jiang, X. (2025). *2025 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, [pdf] 🎧 🧠
- [C9] **Speech Slytherin: Examining the Performance and Efficiency of Mamba for Speech Separation, Recognition, and Synthesis**
Jiang, X., Li, Y. A., Florea, A. N., Han, C., and Mesgarani, N. (2024). *2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, [pdf] [code1] [code2] 🎧 🔧 🧠
- [C8] **StyleTTS-ZS: Efficient High-Quality Zero-Shot Text-to-Speech Synthesis with Distilled Time-Varying Style Diffusion**
 Li, Y. A., Jiang, X., Han, C., and Mesgarani, N. (2024). *The 2025 Annual Conference of the Nations of the Americas Chapter of the ACL (NAACL)*, [pdf] [code] 🎧 🧠
- [C7] **Just ASR+ LLM? A Study on Speech Large Language Models’ Ability to Identify and Understand Speaker in Spoken Dialogue**
Wu, J., Fan, X., Lu, B. R., Jiang, X., Mesgarani, N., Hasegawa-Johnson, M., and Ostendorf, M. (2024). *2024 IEEE Spoken Language Technology Workshop (SLT)*, [pdf] [code] 🎧 🧠
- [C6] **Ssamba: Self-Supervised Audio Representation Learning with Mamba State Space Model**
 Shams, S., Dindar, S. S., Jiang, X., and Mesgarani, N. (2024). *2024 IEEE Spoken Language Technology Workshop (SLT)*, [pdf] [code] 🎧
- [C5] **Style-Talker: Finetuning Audio Language Model and Style-Based Text-to-Speech Model for Fast Spoken Dialogue Generation**
Li, Y. A., Jiang, X., Darefsky, J., Zhu, G., and Mesgarani, N. (2024). *2024 Conference on Language Modeling (COLM)*, [pdf] [code] 🎧 🧠 🧠
- [C4] **Dual-Path Mamba: Short and Long-Term Bidirectional Selective Structured State Space Models for Speech Separation**
Jiang, X., Han, C., and Mesgarani, N. (2024). *2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, [pdf] [code] 🔧
- [J3] **Listen, Chat, and Remix: Text-Guided Soundscape Remixing for Enhanced Auditory Experience**
Jiang, X., Han, C., Li, Y. A., and Mesgarani, N. (2024). *Special Issue on Deep Multimodal Speech Enhancement and Separation, IEEE Journal of Selected Topics in Signal Processing (JSTSP)*, [pdf] 🔧 🧠
- [C3] **Exploring Self-Supervised Contrastive Learning of Spatial Sound Event Representation**
Jiang, X., Han, C., Li, Y. A., and Mesgarani, N. (2023). *2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, [pdf] 🎧
- HiFTNet: A Fast High-Quality Neural Vocoder with Harmonic-plus-Noise Filter and Inverse Short Time Fourier Transform**
 Li, Y. A., Han, C., Jiang, X., and Mesgarani, N. (2023). *arXiv*, [pdf] [code] 🎧 🧠
- [C2] **DeCoR: Defy Knowledge Forgetting by Predicting Earlier Audio Codes**
Jiang, X., Li, Y. A., and Mesgarani, N. (2023). *2023 INTERSPEECH*, [pdf] 🎧

PAPERS	[C1] Phoneme-Level BERT for Enhanced Prosody of Text-to-Speech with Grapheme Predictions Li, Y. A., Han, C., Jiang, X. , and Mesgarani, N. (2022). <i>2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)</i> , [pdf] [code] 🇺🇸🇩🇪	
	[J2] Learning Representations for New Sound Classes with Continual Self-Supervised Learning Wang, Z., Subakan, C., Jiang, X. , Wu, J., Tzinis, E., Ravanelli, M., and Smaragdis, P. (2022). <i>IEEE Signal Processing Letters</i> , [pdf] [code] 🎧	
	[J1] Compute and Memory Efficient Universal Sound Source Separation Tzinis, E., Wang, Z., Jiang, X. , and Smaragdis, P. (2021). <i>Journal of Signal Processing Systems</i> , [pdf] [code] 🔑	
	[Thesis] Vector-Quantized Speech Separation Jiang, X. (2021). <i>Undergraduate Thesis</i> , 🔑	
SKILLS	<u>Languages:</u> English and Mandarin <u>Programming:</u> Python, PyTorch, HuggingFace, SpeechBrain, CUDA, SQL <u>Others:</u> Linux, Azure, AWS, Google Cloud, git, LaTeX	
ACADEMIC SERVICES	<u>Reviewers for</u> <i>IEEE Signal Processing Letters</i> <i>Association for the Advancement of Artificial Intelligence (AAAI)</i> <i>IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP)</i> <i>IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)</i> <i>IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)</i>	
HONORS	• <u>Master of Science Award of Excellence</u> 2024 top 5% GPA in master graduating class • <u>Wei Family Private Foundation Fellowship</u> 2022 - 2024 covering tuitions and stipends for two Ph.D. years • <u>UIUC Bronze Tablet</u> 2021.12 top 3% GPA in undergrad graduating class and defended an undergraduate research thesis • <u>Henry O. Koehler Merit Scholarship</u> 2021 - 2021 for an outstanding scholastic record • <u>Frank C. Mock Scholarship</u> 2020 - 2021 awarded to top ECE students • <u>Daniel W. and Carol A. Dobberpuhl Student Award</u> 2020 - 2021 for an outstanding scholastic record • <u>James Scholar</u> 2019 - 2021 honor program in College of Engineering, UIUC	